Module 5: Introducing Information Extraction

AWS Academy Natural Language Processing

Module overview

- Sections
- Section 1: Information extraction overview
- Section 2: Types of information extraction
- Section 3: Implementing information extraction

- Labs
- Guided Lab: Implementing Information Extraction
- Guided Lab: Working with Entities

Module objectives

- At the end of this module, you should be able to:
- Understand the terms used in information extraction
- Describe the steps for information extraction
- Implement key phrase extraction (KPE)
- Extract entities from text

Section 1: Information extraction overview

Module 5: Introducing Information Extraction

What is information extraction?

- Information extraction (IE) refers to the natural language processing (NLP) task of extracting relevant information from text.
- IE can be a challenge with unstructured data.

IE applications: Digital assistant queries

Where is the nearest coffee shop?

need address

current location

target location

IE applications: Common phrases

Read reviews that mention

easy to install | video card | power supply | great value

triple A titles | ultra settings | liquid cooling | hdmi

ports | gddr6 | overclock | cable management | fps

make sure | installation | every game

IE applications: Tagging news

Trending news topics

sanctions | US Supreme Court | vaccine | Texas

snow storms | climate change | nuclear power | virus

IE applications: Extracting financial data

- Amazon (NASDAQ:AMZN) and Whole Foods Market, Inc. (NASDAQ:WFM) today announced that they have entered into a definitive merger agreement under which Amazon will acquire Whole Foods Market for $42 per share in an all-cash transaction valued at approximately $13.7 billion, including Whole Foods Market's net debt. "Millions of people love Whole Foods Market because they offer the best natural and organic foods, and they make it fun to eat healthy," said Jeff Bezos, Amazon founder and CEO. "Whole Foods Market has been satisfying, delighting and nourishing customers for nearly four decades – they're doing an amazing job and we want that to continue."

Known entity: business

Known entity: person

Relationship

Numeric data

Extracting financial data: Known entities

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Known entity: business | Known entity: person | Relationship | Numeric data

Extracting financial data: Known entities 2

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Extracting financial data: Relationship

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Extracting financial data: Numeric data

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Known entity: business

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Relationship

Numeric data

IE applications: Extracting events

Example event with basic information.

Whoop! Whoop! I just got a **promotion** to manager at AnyCompany!!!

Example event with temporal information.

Hi!
Don’t forget that next **Tuesday, March 9, 2021**, is the new date for the **Charity Ball** at the **Charming Community Center**! See you at **7PM** sharp!

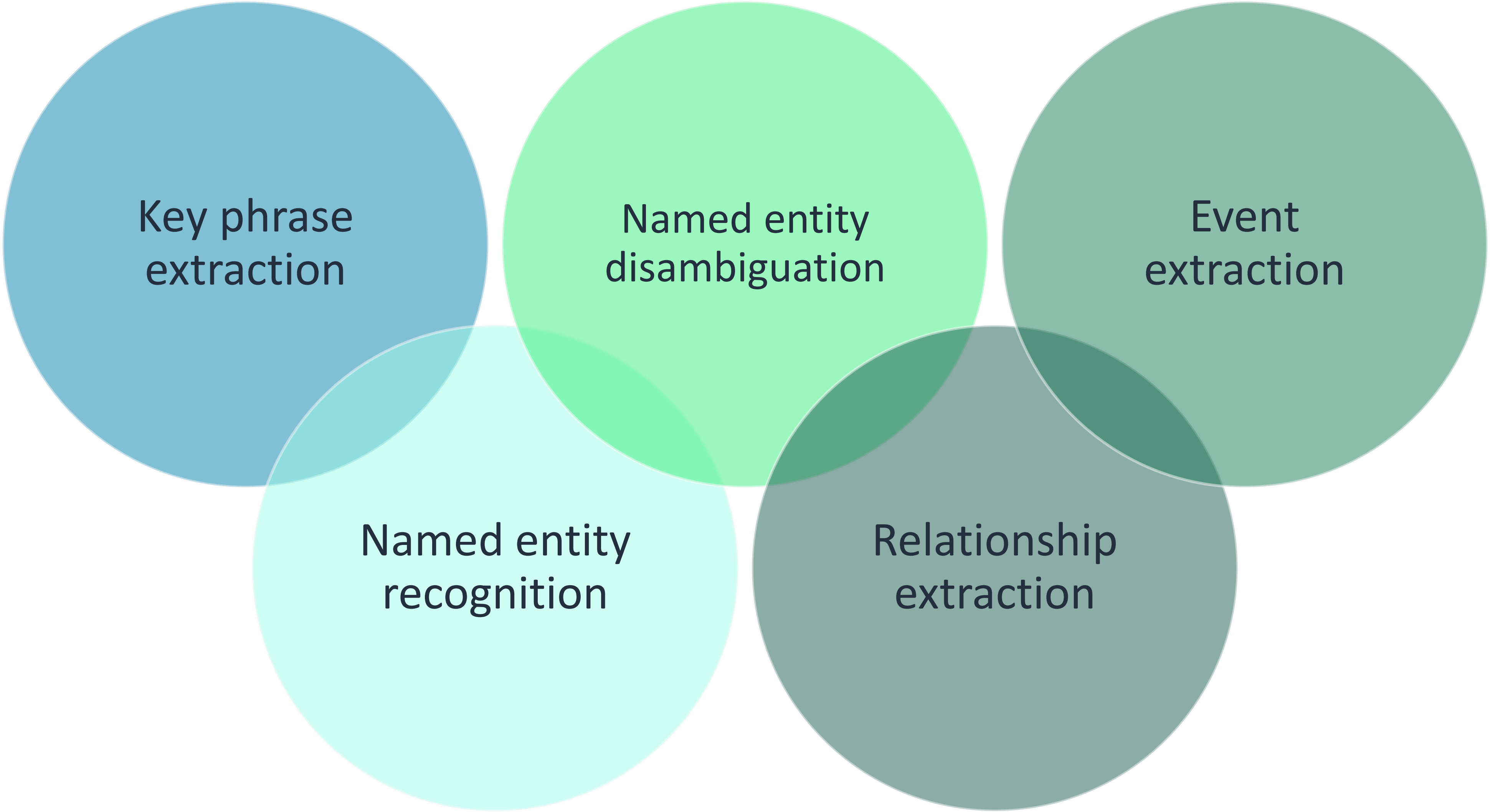
Section 1: Summary

- What is information extraction?
- Information extraction applications
  - Digital assistant extracting information from a question
  - Identifying common phrases or themes in reviews
  - Tagging news
  - Identifying financial data in reports
  - Extracting event information

Section 2: Types of information extraction

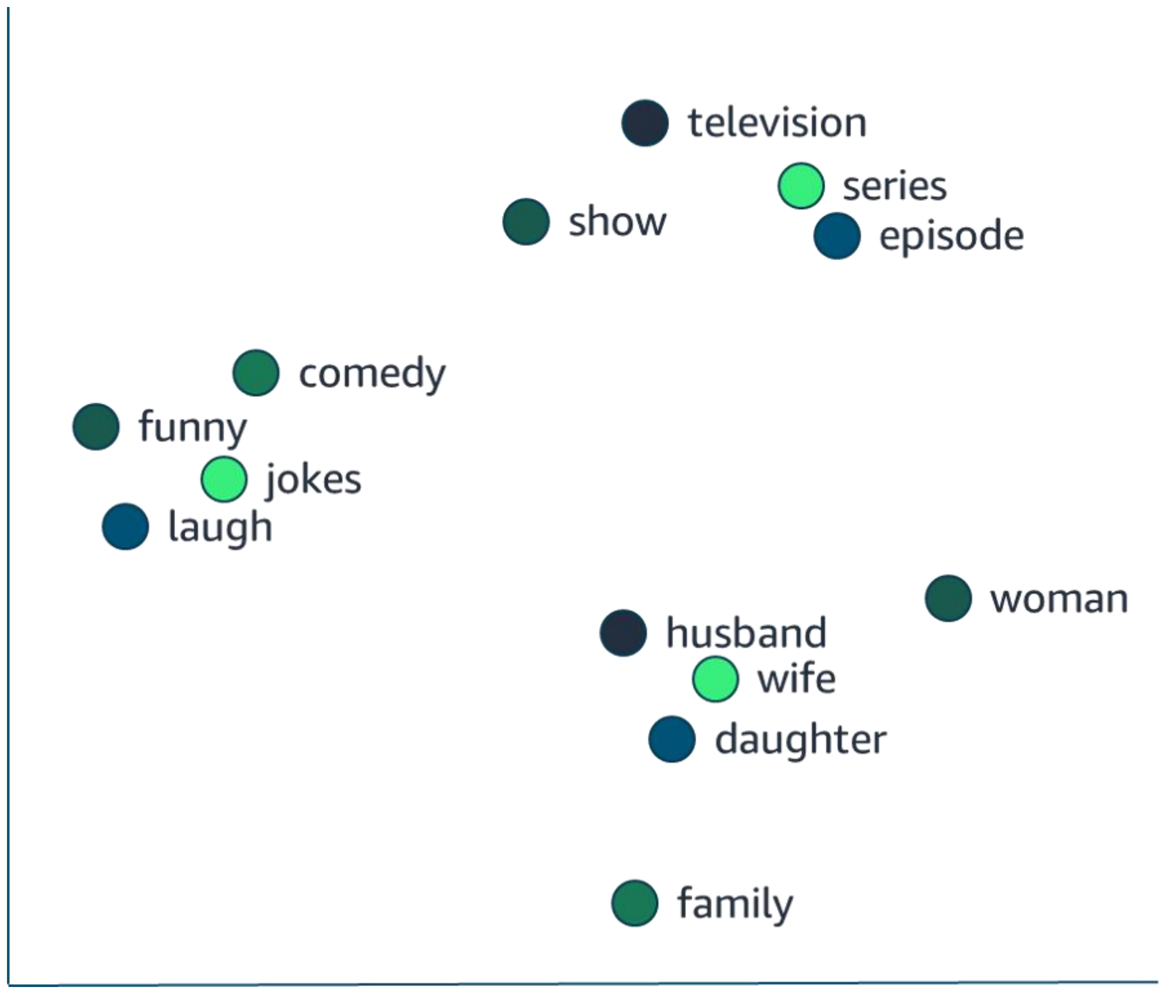
Module 5: Introducing Information Extraction

Types of information extraction



Key phrase extraction (KPE)

- KPE extracts useful words and phrases from text to capture the general intent.
- KPE is generally unsupervised learning.
- Words are nodes in a weighted graph:
    - Weight indicates the importance of a key phrase.
    - How well are the nodes connected with the rest of the graph?
    - The top X nodes are returned.



Determining word significance

- Term frequency-inverse document frequency (TF-IDF) and KeyBERT are used to determine significance of a word.
- Amazon Comprehend provides application programming interface (API) actions for KPE.

Named entity recognition (NER)

- NER is the NLP task of identifying entities such as names, locations, organizations, dates, and product names.
- NER is often needed for downstream tasks such as event or relation extraction and translation.
- spaCy and BERT provide tools for extracting and visualizing named entities
- Amazon Comprehend provides API actions for entity recognition to find insights and relationships in text.

Alexa, where was Alan Turing born?
Which companies were in the news?
Which products are mentioned in the review?

Named entity disambiguation (NED)

- NED is the NLP task of uniquely identifying entities in the text.
  - Ford forded the ford in the Ford.
- NER and NED enable named entity linking (NEL).
  - NEL maps named entities to specific instances in a knowledge base

Relationship extraction

- Relationship extraction is the NLP task that identifies relationships between entities.

“Millions of people love Whole Foods Market because they offer the best natural and organic foods, and they make it fun to eat healthy,” said Jeff Bezos, Amazon founder and CEO.

- How do you extract that Jeff Bezos is the Chief Executive Officer of Amazon?

Event extraction

- Event extraction is the process of identifying and extracting knowledge about events in text data.
- You often use pattern matching and NER in event extraction.

Example event with key information.

> **AnyCompany** announced today a 2-to-1 **stock split**, which will be effective on **May 24, 2021**, at market open.

- NER used to extract AnyCompany
- Pattern matching used to extract the stock split event.
- May 24, 2021 recognized as a date.

Section 2: Summary

- Key phrase extraction
- Named entity recognition
- Named entity disambiguation
- Relationship extraction
- Event extraction

Section 3: Implementing information extraction

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IE with Amazon Comprehend

- Amazon Comprehend uses pretrained models to do the following:
- Detect entities
- Detect events
- Detect key phrases
- Detect personally identifiable information (PII)
- Analyze syntax

Entity detection with Amazon Comprehend

- StartEntitiesDetectionJob: Starts an asynchronous job
- Input and output to Amazon Simple Storage Service (Amazon S3)
- UTF-8 with a max size of 102,400 bytes


```
{
  "File": "small_doc",
  "Entities": [{
    "BeginOffset": 0,
    "EndOffset": 4,
    "Score": 0.645766019821167,
    "Text": "Maat",
    "Type": "PERSON"}]
}
```


location of entity in document

confidence level

text

type

Training custom entity recognizers

- Amazon Comprehend can implement custom entity recognition.
- You can provide data in two ways:

Annotations

Entity lists

Annotations

- Train 25 entity types per model
- Use training data with an annotation file

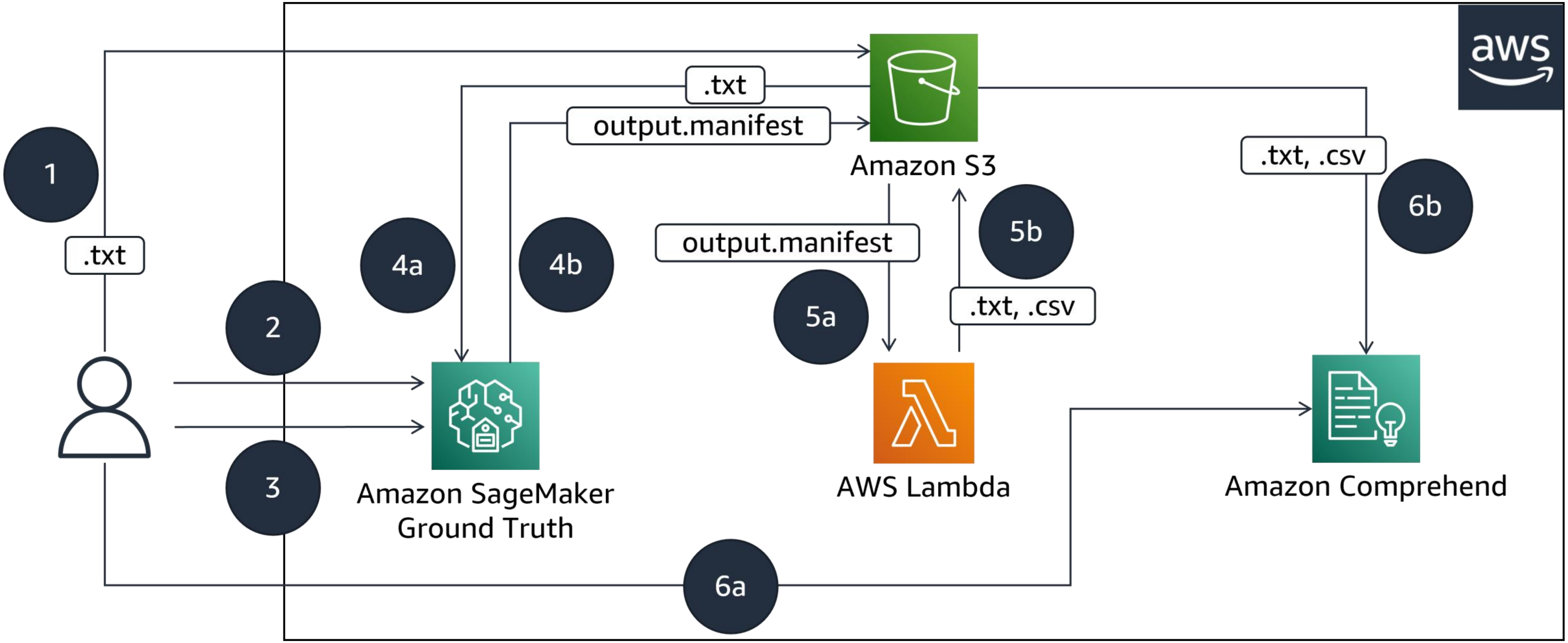
TRAINING DATA

documents.txt
Ana Carolina Silva is an engineer in the high-tech industry.
John Doe has been an engineer for 14 years.
Jorge Souza is a judge on the Washington Supreme Court.
Our latest new employee, Smith, has been a manager in the industry for 4 years.

ANNOTATIONS

File	Line	Begin Offset	End Offset	Type
documents.txt	0	0	18	ENGINEER
documents.txt	1	0	5	ENGINEER
documents.txt	3	25	30	MANAGER

NER with Amazon SageMaker Ground Truth and Amazon Comprehend



Entity lists

- Train 25 entity types per model
- Use training data with an entity list

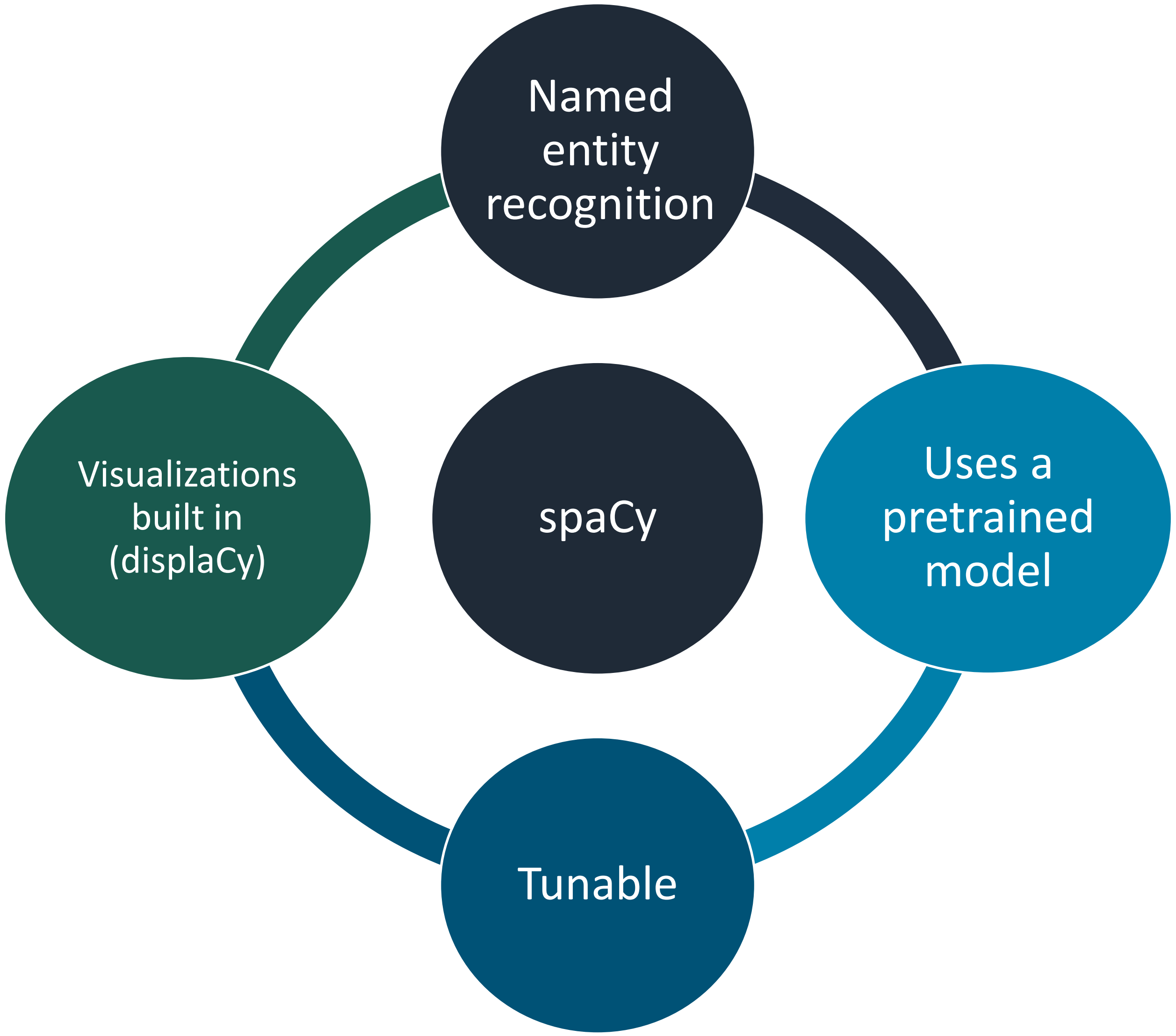
TRAINING DATA

documents.txt
Josef Brown is an engineer in the high-tech industry.
J Doe has been an engineer for 14 years.
Emilio Johnson is a judge on the Washington Supreme Court.
Our latest new employee, Smith, has been a manager in the industry for 4 years.

ENTITY LIST

Text	Type
Jo Brown	ENGINEER
John Doe	ENGINEER
Jane Smith	MANAGER

Extracting information with spaCy



spaCy example

Example code for using spaCy.

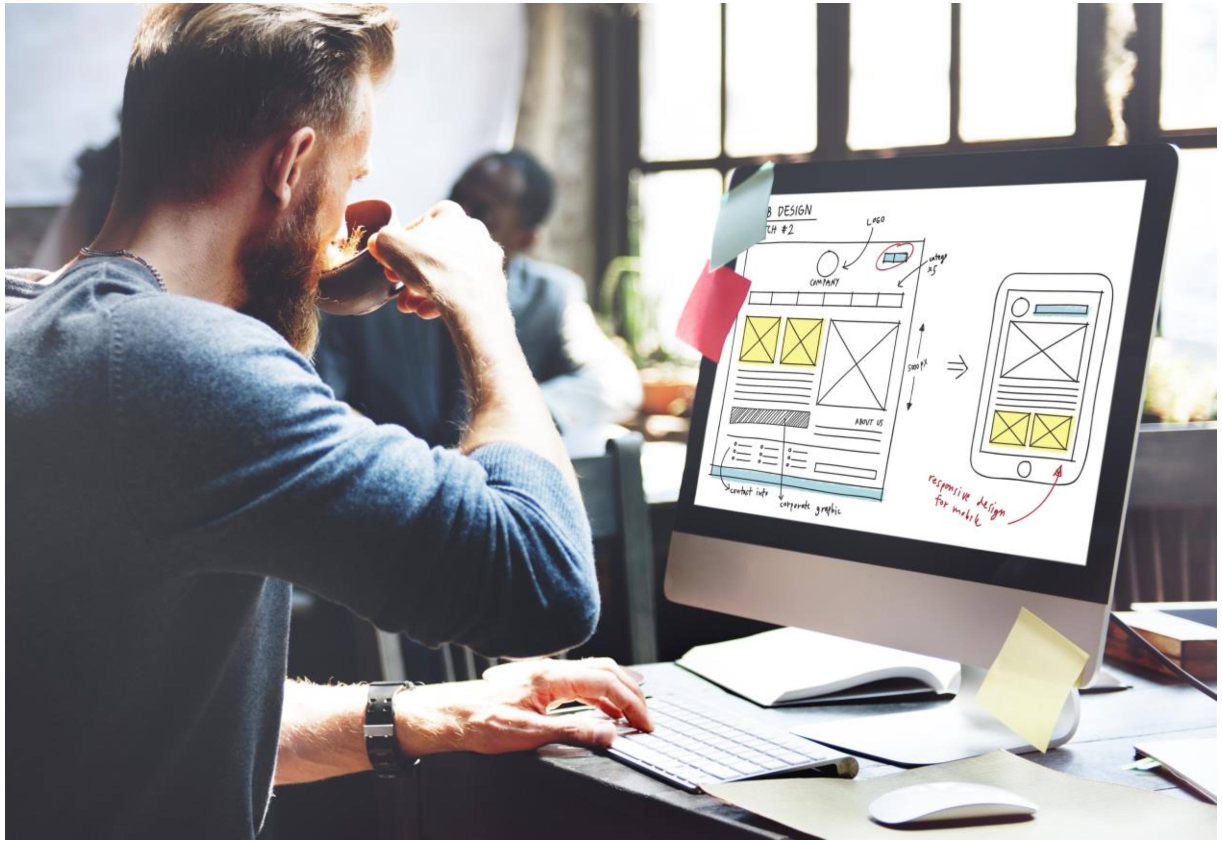
```
import spacy

nlp = spacy.load("en_core_web_sm")
doc = nlp("AnyCompany is looking at buying U.K. startup for $1 billion")

for ent in doc.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

Text	Start	End	Label	Description
AnyCompany	0	10	ORG	Companies, agencies, institutions
U.K.	32	36	GPE	Geopolitical entities, such as countries, cities, and states
$1 billion	49	59	MONEY	Monetary values, including unit

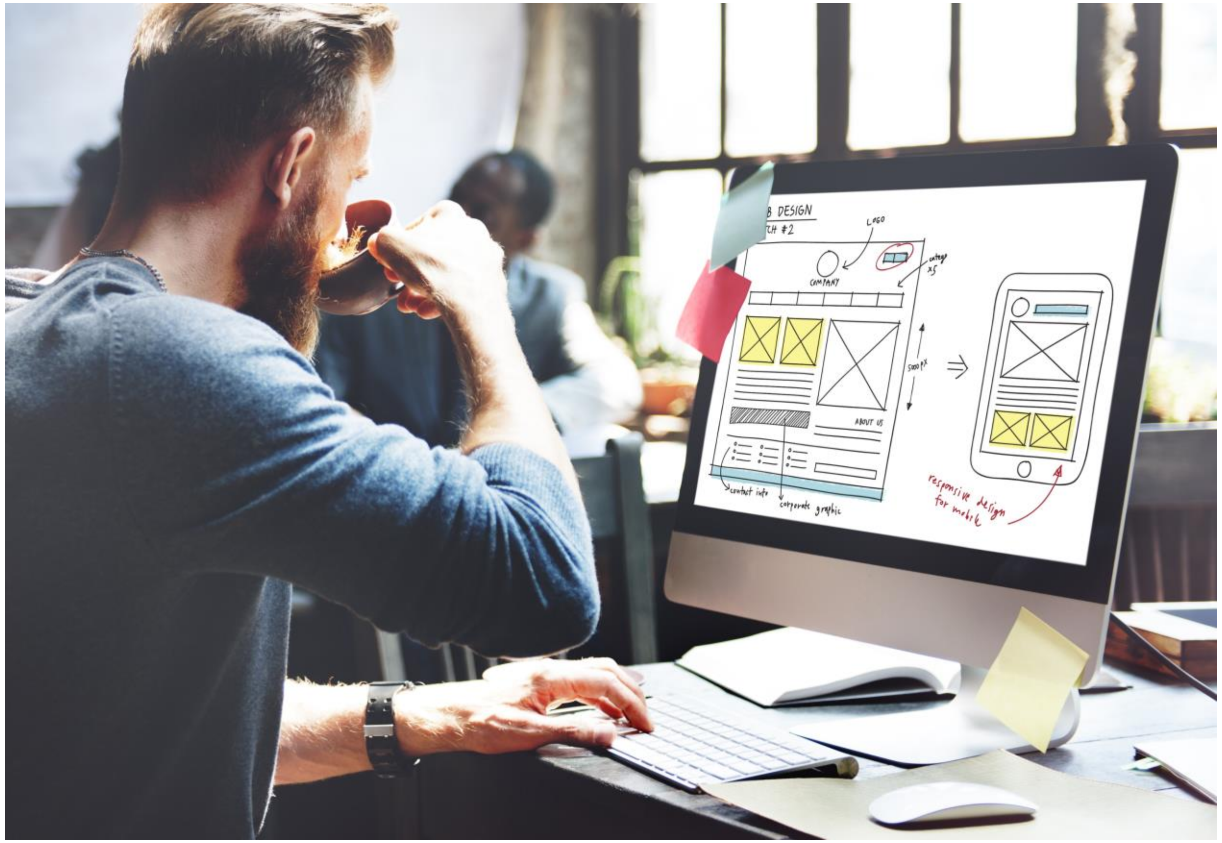
displaCy output:

AnyCompany ORG is looking at buying **U.K. GPE** startup for **$1 billion MONEY**

Module 5 Guided Lab 1: Implementing Information Extraction



Module 5 Guided Lab 2: Working with Entities



Section 3: Summary

- Amazon Comprehend
  - Pretrained models
  - Custom models
- spaCy
  - Entity recognition system
  - Assigns labels to contiguous spans of tokens

Thank you

Corrections, feedback, or other questions?
Contact us at https://support.aws.amazon.com/#/contacts/aws-academy.